Datasheet

RGB-621-R1 RGB Lighting Control Module



Part No.: RGB-621-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board)

Controller: RP2350A

Functional Overview

The RGB-621-R1 is a DIN-rail mounted RGBCCT lighting control module for 12-24 V DC LED systems. It provides five low-side MOSFET LED outputs (R, G, B, CW, WW), two 24 V digital inputs, one relay output, RS-485 Modbus RTU communication, and USB-C configuration interface.

Board	Function
MCU Board	RP2350A controller, RS-485 communication, USB-C interface, power conversion
Field Board	MOSFET LED outputs, Relays, DI

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (32 Mbit)
Power Input	24 V DC nominal
Input Protection (24 V logic)	1 A fuse, reverse polarity diode (STPS340U), TVS surge suppression, EMI filtering
LED PS input protection	10 A fuse; TVS surge suppression; no reverse-polarity protection on this input
Main Logic Supply	Buck regulator 24 V → 5 V, 3.3 V LDO regulator
Document revision	DS-RGB-621-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
LED Outputs	5 × low-side MOSFET outputs (AP9990GH-HF: R, G, B, CW, WW)
LED Output Voltage	External 12–24 V DC supply
Maximum Total LED Current	10 A, shared across all channels in use (10 A fuse on LED PS input)
Any single PWM channel	10 A — same shared budget as the module; no per-channel ceiling below 10 A
Relay outputs	1 × relay, NO and C only (SPST-NO); 3 A @ 250 VAC (module limit)
Digital Inputs	2 × IEC 61131-2 compliant 24 V digital inputs (ISO1212 input receiver, module-wetted; not isolated); per-channel PTC fuse, TVS, reverse-polarity protection
User Interface	2 buttons, 8 LEDs (power, 2 user, RX, TX, 1× relay, 2× DI)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C, 5 V logic, ESD protected
Typical Power Consumption	1.85 W typical / 3.0 W maximum (no LED load)
Modbus defaults	Address 3, 19200 baud, 8N1

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Typical module power consumption: 1.85 W (no LED load).

Maximum module power consumption: 3.0 W (no LED load).

Top Terminal Block (1–9)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	+	LED_PS	LED PSU positive (12 or 24 V)
2	-	LED_PS	LED PSU negative
3	V+	POWER	24 V DC input
4	0V	POWER	24 V DC return
5	NO	RELAY	Relay normally open
6	C	RELAY	Relay common
7	I1	DI	Input 1
8	GND	DI	Input return
9	I2	DI	Input 2

Bottom Terminal Block (1–9)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	+	LED_OUT	Common anode for all channels
2	R	LED_OUT	Red channel
3	G	LED_OUT	Green channel
4	B	LED_OUT	Blue channel
5	CW	LED_OUT	Cool white channel
6	WW	LED_OUT	Warm white channel
7	COM	RS485	RS-485 signal reference
8	A	RS485	RS-485 data +
9	B	RS485	RS-485 data -

Installation, Environmental & Mechanical

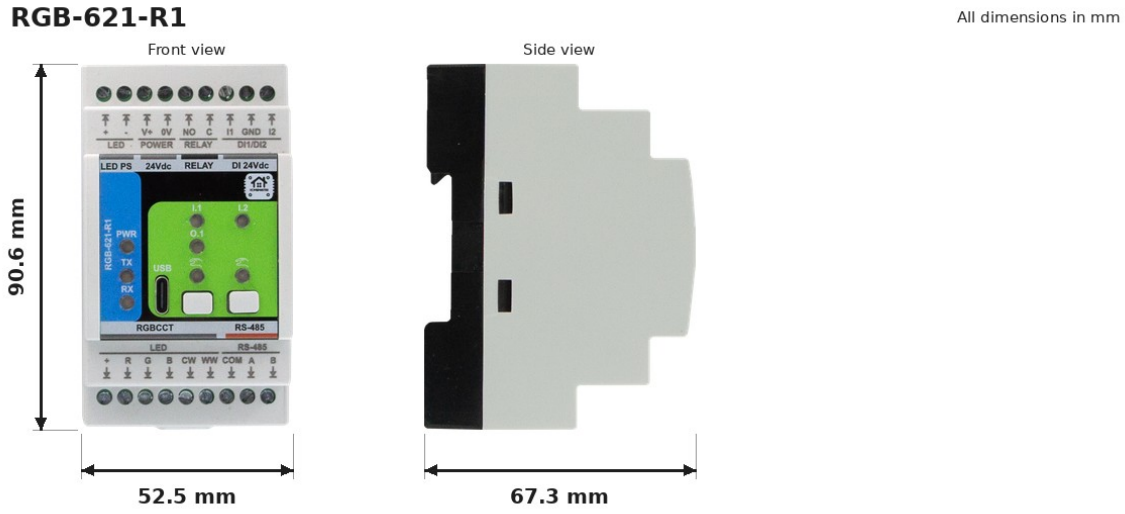
Specification	Details
Terminal Type	Pluggable screw terminal blocks DB129V-5.08 / DB129R-5.08, 5.08 mm pitch
Wire Cross Section	0.2–2.5 mm^2 (AWG 24–12)
Tightening Torque	0.4–0.6 Nm

Specification	Details
Operating temperature	0 °C to +40 °C
Storage temperature	−10 °C to +55 °C
Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m
Pollution degree	2

Specification	Details
Dimensions	52.5 × 90.6 × 67.3 mm (L × W × H)
DIN width	3 modules (≈ 52.5 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks DB129V-5.08 / DB129R-5.08, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	150 g
Gross weight	240 g
Pack size	140 × 125 × 94 mm (L × W × H)
Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm
Notes	All dimensions shown in millimeters unless stated otherwise

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

This section applies to RS-485. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Examples: J-Y(ST)Y 2×2×0.5 mm² or LI2YCY PiMF 2×2×0.50

Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

Compliance & Certifications

The RGB-621-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

Safety Notice

The power input and signal terminals are SELV circuits.

Installation by qualified personnel only.

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